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ORIGINAL ARTICLE

'MN1606SP' by 'Spencer' filial soybean population reveals novel quantitative trait loci and interactions among loci conditioning SDS resistance

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Abstract

Key message **Four novel QTL and interactions among QTL were identified in this research, using as a parent line the most SDS-resistant genotype within soybean cultivars of the US early maturity groups.**

Abstract Soybean sudden death syndrome (SDS) reduces soybean yield in most of the growing areas of the world. The causal agent of SDS, soilborne fungus *Fusarium virguliforme* (Fv), releases phytotoxins taken up by the plant to produce chlorosis and necrosis in the leaves. Planting resistant cultivars is the most successful management practice to control the disease. The objective of this study was to identify quantitative trait loci (QTL) associated with the resistance response of MN1606SP to SDS. A mapping population of $F_{2:3}$ lines created by crossing the highly resistant cultivar 'MN1606SP' and the susceptible cultivar 'Spencer' was phenotyped in the greenhouse at three different planting times, each with three replications. Plants were artificially inoculated using SDS infested sorghum homogeneously mixed with the soil. Data were collected on three disease criteria, foliar disease incidence (DI), foliar leaf scorch disease severity (DS), and root rot severity. Disease index (DX) was calculated as DI × DS. Ten QTL were identified for the different disease assessment criteria, three for DI, four for DX, and three for root rot severity. Three QTL identified for root rot severity and one QTL for disease incidence are considered novel, since no previous reports related to these QTL are available. Among QTL, two interactions were detected between four different QTL. The interactions suggest that resistance to SDS is not only dependent on additive gene effects. The novel QTL and the interactions observed in this study will be useful to soybean breeders for improvement of SDS resistance in soybean germplasm.

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Introduction

Sudden death syndrome (SDS), a soilborne disease of soybean [*Glycine max* (L.) Merr.] in the US caused by the fungus *Fusarium virguliforme* (Fv), is an important yield deterrent. In 2014, losses were estimated at 1.7 million metric tons, ranking the disease as the second worst in the United States (Bradley and Allen 2014). Yield losses from 5 to 15%, and up to 80% have also been reported (Roy et al. 1997), depending on the prevalent environmental conditions (Leandro et al. 2013). The fungus colonizes the soybean root causing a rot, identified by brown to black discoloration in the excised roots (Roy 1997). Once the mycelium colonizes the root and hyphae penetrate the xylem (Navi and Yang 2008), a phytotoxin(s) produced by the fungus is translocated to the leaves causing chlorotic and necrotic symptoms (Brar et al. 2011; Jin et al. 1996; Chang et al. 2016). The fungus, however, has never been isolated from the leaf tissue.

The field management of the Fv causing disease is difficult. The production of chlamydospores by *F. virguliforme* and its wide host range make crop rotation a somewhat ineffective management strategy (Kolander et al. 2012; Marburger et al. 2014). The results of using fungicides as seed treatments have been variable, some having no effect

(Weems et al. 2015; Chilvers et al. 2011), while others reduce the symptoms (Mueller et al. 2011; Kandel et al. 2016). It has been shown that delaying planting to avoid cold and wet soils is a strategy to disfavor SDS disease development (Leandro et al. 2013). The planting of resistant cultivars is, however, the most successful strategy. Unfortunately, no commercial cultivars showing complete resistance to Fv and to the disease SDS are available.

Breeding and development of Fv-resistant cultivars is an involved process due to the quantitative inheritance of the SDS disease resistance trait. In four populations of maturity groups IV and later, and applying SSR, RAPD, RFLP, or SNP molecular markers, 75 marker trait associations were listed (de Farias-Neto et al. 2007; Hnetkovsky et al. 1996; Luckew et al. 2013; Meksem et al. 1999; Iqbal et al. 2001; Njiti et al. 2002; Kassem et al. 2006, 2007, 2012; Kazi et al. 2008; Prabhu et al. 1999; Srour et al. 2012), on SoyBase: the Soybean Breeder's Toolbox surveyed on April 21, 2017.

Additional QTL associated with resistance to SDS had been identified although not included in the SoyBase data checked on April 21, 2017. Recently, in the population of MD96-5722 × 'Spencer', 12 additional QTL were mapped with SNP markers (Anderson et al. 2015). Similarly in 2014, a genome-wide association study identified 13 additional QTL (Wen et al. 2014), and two more QTL were also identified in a study in which phenotyping was done using the stem cutting assay as the screening protocol (Swaminathan et al. 2015). Across all available studies, the R^2 values for the QTL ranged from 0.01% (Anderson et al. 2015) to 47% (Njiti et al. 1998).

The need for identifying novel sources of major QTL for SDS disease resistance in soybean continues. The disease has spread into the northern central soybean production region of the U.S. (Kurle et al. 2006; Chilvers and Brown-Rytlewski 2010; Tande et al. 2014) and even into Southern Canada (Anderson and Tenuta 1998), making the search for genotypes of the maturity groups (MG) III and earlier with SDS resistance an important need.

The MG I food specialty cultivar 'MN1606SP', developed by the University of Minnesota soybean breeding program in 1994 (Personal communication, Dr. Orf, 2014), is one of the most SDS-resistant genotypes available in the Southern Illinois University at Carbondale's commercial SDS variety field tests (Personal communication, Cathy Schmidt 2005). The results were consistent and repeatedly observed since that time although MN1606 was never screened nor purposely selected for SDS disease resistance during its development. The objective of this study was therefore to map QTL from this highly SDS disease resistant early-maturing cultivar. This mapping effort is one of the first single cross studies conducted on early maturing germplasm adapted to the northern soybean production region of the US. To conduct the study, a mapping population was developed by crossing MN1606SP to the susceptible cultivar 'Spencer' (Wilcox et al. 1989). A modified SDS screening protocol developed by Luckew et al. (2012) using an increased concentration of SDS inoculum was used as a means to differentiate QTL that could have minor effects on the final expression of resistance, thus providing a more efficient phenotyping tool than protocols previously used.

Materials and methods

Plant materials

For the study, 200 $F_{2:3}$ derived lines from the cross between 'MN1606SP' and 'Spencer' (Wilcox et al. 1989) were used. The resistant parent 'MN1606SP' was developed by Dr. James Orf at the University of Minnesota by crossing M90-764 × M90-2144. The experimental line M90-764 was developed from the cross of ['Evans' × ('Anoka' × 'Amsoy')] × ['Peterson 0877′ × ('Clay' × 'Evans')]. The experimental line M90-2144 originated from the cross of A85-193023 × 'Kato' (Personal communication, Dr. Orf, 2014). The cultivar MN1606SP is resistant to SDS, however, there is no information available to determine from which of the ancestor lines the resistance to SDS was derived. 'MN1606SP' was first identified as SDS resistant in the Southern Illinois University at Carbondale's commercial SDS variety field tests in 2005 (Personal communication, Cathy Schmidt 2005).

The mapping population of 200 $F_{2:3}$ lines was developed at the Iowa State University soybean research site located at the Puerto Rico Agricultural Experiment Station, University of Puerto Rico, Isabela, Puerto Rico. The F_1 seeds were sown on July 2012 and each F_1 plant was harvested individually. During the growing season, the hybrid nature of the F_1 plants was confirmed on the basis of morphological markers. The F_1 plant identity was maintained during the subsequent generation advance. F_2 plants (F_3 seed generation), were individually harvested in October 2013 to develop $F_{2:3}$ lines. A total of 200 individual $F_{2:3}$ lines with sufficient seed numbers to conduct replicated tests were used in the study.

Phenotyping SDS response

Inoculum preparation and screening medium

Inoculum was produced and lines were screened for reactions to SDS according to Luckew et al. (2012). The *Fusarium virguliforme* isolate LL0009, also identified as NE305, was obtained in 2006 from a symptomatic plant in Nevada, IA, and used in all runs. The isolate was grown on 1/3 strength potato dextrose agar for one month on a lab bench at room temperature exposed to 24-h 40-W fluorescent light. Two liter Mason jars were filled with 750 ml of white milo sorghum [*Sorghum bicolor* (L.) Moench] seeds and the jars were filled with water to soak the seeds for 24 h. Afterward, the water was poured out, the jars were autoclaved for 1 h on two consecutive days and inoculated with five mycelia plugs (7 mm diameter), of the *F. virguliforme* culture. The jars were shaken daily during 1 min, kept on a lab bench at room temperature and exposed to 24-h 40 W fluorescent light for 2 weeks. In previous experiments (Luckew et al. 2012), qPCR was used to measure the *F. virguliforme* DNA/mg of infested sorghum seed prepared for each run, which ranged from 21.4 to 31.1 ng dry weight. The experiment for the present study was run three times with new inoculum prepared following the procedure described for each of the runs.

A soil mixture consisting of a 1:1 ratio (v/v) of soil and sand was used as the greenhouse media. The inoculum was homogeneously mixed by hand with the soil mixture in a 1 part inoculum to 20 parts soil mix (v/v). Negative controls containing sterile non-infested sorghum seed were used with the resistant cultivar Ripley (Cooper et al. 1990) and the susceptible cultivar Essex (Smith and Camper 1973). These cultivars were also used as positive controls. The non-infested sorghum seed plantings were used to confirm that the original sorghum seed used was not contaminated.

Phenotypic data recording

Foliar disease symptom assessment was done 36 dap, at soybean plant growth stage V2–V3 (Fehr and Caviness 1977). Two visual foliar phenotypic SDS disease scores were determined, disease incidence (DI), and disease severity (DS), and the third, disease index (DX) was calculated.

Disease incidence was the number of plants showing foliar disease symptoms divided by the total number of living plants in the cup. The plants showing SDS foliar leaf chlorosis and necrosis symptoms as described by Roy et al. (1997) were scored together as an average for disease severity (DS) using a percentage scale from 0 to 100 in unit increments of 5. The plants not showing symptoms were excluded from the cup averages for DS. Disease incidence and DS were used to calculate DX as

$$\mathrm{DX} = \mathrm{DS} \times \mathrm{DI} \div (\text{the maximum value DS can be on the scale used}).$$

For this research, the maximum value of the scale is 100. Since DI is multiplied by 100 to obtain a percentage, the mathematical equation reduces to DS × DI.

At 37 dap, plants were carefully removed from the cups with the root ball intact and the roots were washed in warm water. The root area showing dark brown to black discoloration (Roy et al. 1997) was visually assessed in a percentage scale from 0 to 100 in increments of 5 of the total root area. Root rot severity was the fourth disease criteria recorded.

Experimental conditions and design

A plot was considered to be five seeds of each genotype planted into a single 240-ml Styrofoam cup at a depth of 2 cm. The cups were placed in a greenhouse room in a randomized complete block design.

The room was maintained at 23.6 ± 5 °C with a 16-h photoperiod. According to a sensor on the roof of the greenhouse, light intensity ranged from 836 w/m^2 during peak sunlight to 167 w/m^2 in the mornings and evenings. For the duration of the experiment except on day 21 after planting, each cup was watered once daily to maintain soil moisture. To simulate a flood-like event, on day 21 after planting, the cups were watered twice during the day, once in the morning and again in the afternoon. This date was chosen due to this being the first day symptoms usually appear under greenhouse conditions. Afterwards the watering was done only once a day, until the end of the experiment that occurred 37 days after planting.

The experiment consisted of three runs, with each genotype planted in three different cups and each cup was considered a replication within the run. The first run was planted in the ISU Plant Pathology Greenhouses, Ames, IA on December 10, 2013. The second run was planted in the ISU Agronomy Greenhouses, Ames, IA planted on December 11, 2013. The third run was planted on February 17, 2014 in the ISU Agronomy Greenhouses, Ames, IA. In each replication/run, genotypes were the 200 $F_{2:3}$ lines, the parents of the population and the controls.

Genotyping and linkage map construction

Genomic DNA was collected from leaf samples of 12 individual plants per genotype at 36 dap. To ensure sufficient amounts of DNA, leaf samples were collected using two of the three runs conducted. The DNA from each run was isolated separately following the 'Small scale extraction of high quality DNA' protocol (CIMMYT 2005). The DNA was quantified by running 2 µl on an agarose gel with a New England Biolabs 1 kb ladder. The final number of $F_{2:3}$ lines used was 190 as 10 of the 200 lines had extracted DNA that was of poor quality. The DNA samples of the 190 $F_{2:3}$ lines and that of the parents were sent to Dr. Cregan's lab (USDA ARS, Beltsville, MD) for the Illumina (https://www.illumina.com) Infinium Genotyping assay. The BACRSoySNP6K BeadChips (Song et al. 2013) were used for the genotyping assay.

R/QTL (R Core Team 2015; Broman et al. 2003) was used to generate the genetic linkage map following the guide found on the r/qtl website: http://www.rqtl.org/tutorials/geneticmaps.pdf (Browman 2010). Of the 5403 markers, 147 were missing genotypic data for all the lines and those were removed. The remaining 5256 markers were then observed for homozygosity. There were 3698 non-segregating homozygous markers removed from the 5256, leaving 1558 markers. After cleaning the data set for missing data, and setting the minor allele frequency to 0.05, the 1558 SNP loci were reduced to 1326. The genetic map created from these 1326 SNPs covered 2636.7 cM with an average distance of 2.0 cM between markers on 20 linkage groups (LG) (Table 1) using the Kosambi mapping function.

Statistical and QTL analysis

Normality tests and ANOVA tests were conducted using SAS version 9.3 (SAS Institute 2011). Main effects tested were run, replication, genotype, and interactions. All effects were considered random. The MS for genotypes in the ANOVA test were apportioned to the $F_{2:3}$ lines and the parents. No significant interactions between run and genotype for each of the three foliar disease assessments and root rot severity were detected; therefore, the interaction terms were pooled into the error term. To test for the presence of transgressive segregants, contrast statements in PROC GLM were used. Broad sense heritability was calculated using the mean squares of the main effects from the ANOVA table. A distribution with skewness of 0 and a kurtosis of 3 was considered normal according to Pearson (1895).

Table 1 Breakdown of the genetic map by chromosome of the 'MN1606SP' by 'Spencer' $F_{2:3}$ derived lines

Chromosome no.	Coverage (cM)	No. of markers	cM per marker
Chr_1	87.2	48	1.8
Chr_2	160.7	44	3.7
Chr_3	128.5	50	2.6
Chr_4	148.2	56	2.7
Chr_5	150.0	44	3.4
Chr_6	147.5	53	2.8
Chr_7	102.3	72	1.4
Chr_8	151.0	104	1.5
Chr_9	111.5	48	2.3
Chr_10	144.4	89	1.6
Chr_11	113.1	19	6.0
Chr_12	119.5	82	1.5
Chr_13	185.3	56	3.3
Chr_14	83.5	62	1.4
Chr_15	124.3	115	1.1
Chr_16	140.2	71	2.0
Chr_17	157.9	73	2.2
Chr_18	131.9	67	2.0
Chr_19	105.3	69	1.5
Chr_20	144.3	104	1.4
Total	2636.6	1326	2.0

QTL mapping was performed using the composite interval mapping method with a window size of 10 cM and five markers as covariates using the function 'CIM' in R/QTL (R Core Team 2015; Broman et al. 2003). The interactions and estimated effects were obtained using the addint() and fitqtl() functions in R/QTL. To determine the experiment wise error of 0.05, one thousand permutations were run to estimate the critical LOD threshold values which were 3.99, 4.03 and 3.97 for DI, DX, and root rot severity, respectively. The DS QTL mapped to regions that overlapped with intervals that DX QTL mapped and none were novel. Since DX refers to all plants in the plot rather than individual plant data it was decided not to present the DS QTL results.

Results

Distributions of phenotypic data

According to a Shapiro–Wilk test for normality, both DI ($p = 0.38$), and root rot ($p = 0.66$) follow normal distribution, while DS ($p < 0.006$) and DX did not ($p < 0.002$). Considering skewness and kurtosis, both DS and DX are <1.00 for skewness with a normal curve having a skewness of 0 and within 1.5 of the standard value of 3 for kurtosis (Fig. 1).

There was a significant effect of genotype ($p < 0.05$) on all phenotypes when partitioning out the progeny (Table 2). No significant differences were detected for the interaction term of run × genotypes (data not shown). On this basis, the phenotypic observations of each genotype were combined across runs. For the three foliar disease assessments, DI, DS, DX and the root rot severity data, 16% of the lines had scores that were classified as resistant, and similar to the resistant parent 'MN1606SP' (data not shown). For root rot severity, one of the lines, however, was significantly ($p = 0.0219$) more resistant than 'MN1606SP', indicating transgressive segregation for the trait.

The broad sense heritability for foliar symptoms were 0.26 for DI, 0.24 for DS, 0.21 for DX, and 0.23 for root rot severity, lower than previously reported (Anderson et al. 2015; Santichon et al. 2004). In the studies, the SDS disease was assessed under different environments, which may explain the difference in heritability values. It has been established that the expression of the SDS is highly dependent on environmental conditions (Leandro et al. 2013).

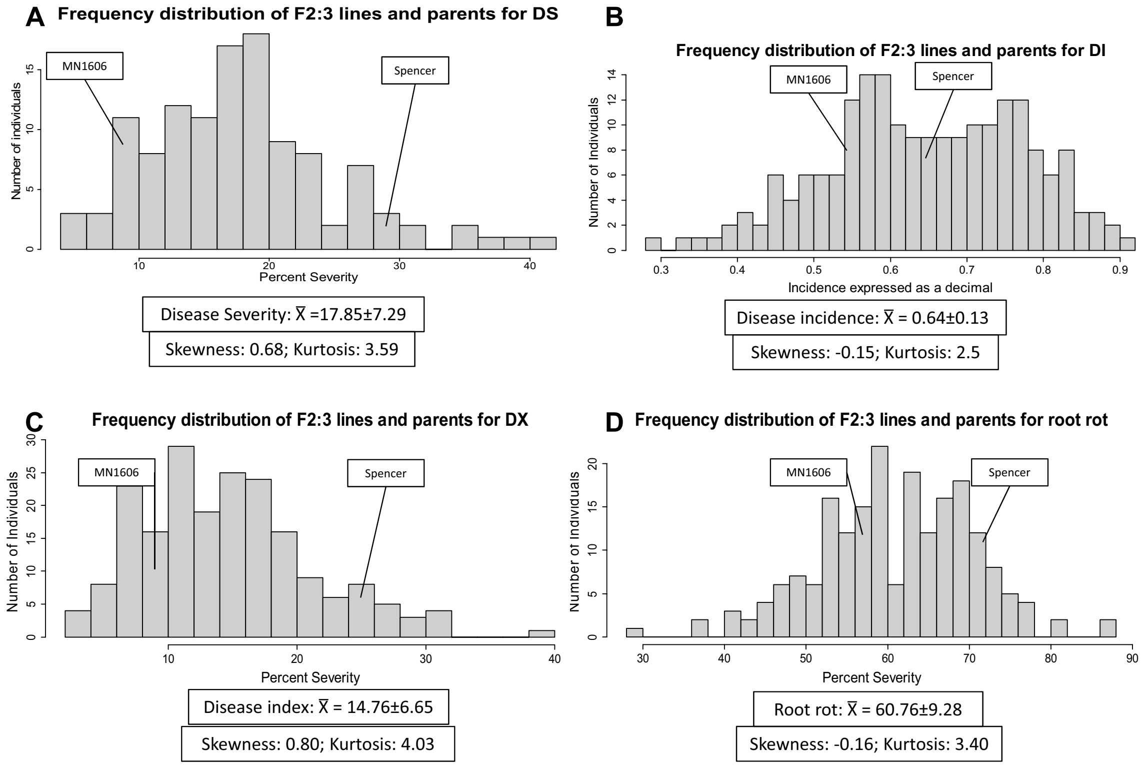

Fig. 1 Frequency distributions of the means from three runs of **a** germination rate, **b** disease incidence, **c** disease index, and **d** root rot of the $F_{2:3}$ RILs from the cross MN1606' by 'Spencer'. *Arrows* indicate the parents. The mean and standard deviation (SD) (mean ± SD), skewness and kurtosis are shown below the distributions

Identification of QTL

A total of ten QTL were identified across seven of the 20 chromosomes in the segregating progeny of the cross of 'MN1606SP' × 'Spencer' (Table 3; Fig. 2). A QTL was considered novel if the region identified in this study is beyond 15 cM of a previously identified QTL for the same disease metric used. The estimated additive effect values are based on the additive effect of an allele substitution for the QTL based on each disease metric, DI, DX, and root rot. Positive values mean that the allele from MN1606SP provides greater resistance than the Spencer allele. Three QTL were identified for DI (Table 3; Fig. 2) on three chromosomes, 13 (LG F), 15 (LG E), and 16 (LG J). On chromosome 13, the peak LOD score for the QTL was 4.4 and it was found between 71 and 124 cM. It explained 11.4% of the variation, and had an estimated additive effect of -0.1. The QTL on chromosome 15 found between 39 and 48 cM, had a peak LOD of 4.3, explained 9.7% of the variation and had an estimated additive effect of 0.1. Chromosome 16 had a DI QTL with a LOD score of 4.9, located between 63 and 80 cM contributing 11.8% of the total variation and had an estimated additive effect of 0.1. The QTL identified on chromosome 16, DI3, is considered novel.

Four QTL for DX were found on four chromosomes (Table 3; Fig. 2). The QTL accounting for the largest percent variation was found on chromosome 14 (LG B2) between 41 and 68 cM with a LOD score of 13.2, explaining 34.4% of the variation with an estimated additive effect of 3.9. A DX QTL was identified in the same 2 LOD confidence interval as the DI QTL (39–48 cM) on chromosome 15. For the DX QTL in this interval, the LOD score was 5.8 accounting for 13.0% of the variation with an estimated additive effect of 3.3. The QTL on chromosome 4 (LG C1) between 81 and 85 cM had the lowest LOD score of the DX QTL at 4.1, explaining 9.0% of the variation, with an additive effect of 0.4, and a dominance effect of −4.4 (data not shown in Table 3). Another QTL found in the analysis for DX was on chromosome 20 (LG I) between 28 and 36 cM with a LOD of 8.5 explaining 20.0% of the variation with an estimated additive effect of −0.5.

Three QTL were identified for root rot severity located on two different chromosomes, one on chromosome 14, and two on chromosome 6 (LG C2), (Table 3; Fig. 2). All three of these QTL are considered novel. The QTL on chromosome 14 found in the interval between 68 and 88 cM had a LOD score of 9.2 and explained 26.8% of the variation with an estimated additive effect of 0.6. Chromosome 6 had two QTL 70 cM apart, the first being in a 2 LOD interval of

Table 2 ANOVA table for the four disease assessment criteria: disease incidence (DI), disease severity (DS), disease index (DX), and root rot severity

Main effect	*df*	MS	*F* value	Pr > *F*
Disease incidence (CV: 50.40)				
Run	2	29.05	273.19	<0.0001
Replication (run)	6	1.23	11.59	<0.0001
Genotype	199	0.13	1.25	0.0158
Progeny	197	0.13	1.25	0.0136
Parental	1	0.04	0.42	0.5152
Disease severity (CV:104.62)				
Run	2	22,731.69	67.36	<0.0001
Replication (run)	6	1911.93	5.67	<0.0001
Genotype	199	441.43	1.31	0.0042
Progeny	197	437.54	1.30	0.0057
Parental	1	1612.72	4.78	0.0290
Disease index (CV:115.59)				
Run	2	27,384.36	95.29	<0.0001
Replication (run)	6	1577.05	5.49	<0.0001
Genotype	199	367.22	1.28	0.0082
Progeny	197	366.26	1.27	0.0091
Parental	1	904.89	3.15	0.0762
Root rot severity (CV: 41.26)				
Run	2	125,299.43	196.86	<0.0001
Replication (run)	6	5753.65	9.04	<0.0001
Genotype	199	769.45	1.21	0.0322
Progeny	197	770.46	1.21	0.0319
Parental	1	1146.74	1.80	0.1797

Replication was nested within run

3–22 cM with a LOD score of 5.7 explaining 15.6% of the variation and an estimated additive effect of 0.8. The second, between 90 and 98 cM, had a LOD of 6.5 explaining 18.0% of the variation with an estimated additive effect of 1.7.

For each disease criteria, a multivariate model was constructed that included all the QTL associated with it and their interactions (Table 3). Each model was run through fitqtl() on *R*/QTL, and the R^2 values were calculated. The percent variation explained by the models for each disease criteria was 36.1% for DI, 14.0% for DX, and 37.0% for the root rot severity.

Identification of QTL interactions

The addint() analysis identified an additive by additive interaction between the QTL associated with root rot severity on chromosome 14, Rot2, and that on chromosome 6 that was in the interval between 3 and 22 cM, Rot3. Each region individually has additive effects of 0.6 and 0.8 located in chromosomes 14 and 6, respectively. When both regions are homozygous for the 'MN1606' allele, however, this interaction had an estimated additive by additive effect of −8.7 with a LOD score of 5.4, explaining 14.7% of the variation (Table 3; Fig. 3a).

While the QTL on chromosome 20 for DX has a relatively small additive effect of −0.5, it was shown to have a statistically significant interaction with the QTL on chromosome 14 associated with DX, with a LOD 8.0 explaining 18.8% of the variation (Table 3; Fig. 3b). The interaction is additive by dominance with an estimated joint effect of 8.0. When the QTL on chromosome 14 is heterozygous, it interacts with the 'Spencer' allele to increase disease index (Fig. 3b). The heterozygosity level in the random $F_{2:3}$ mapping population allowed for this interaction to be identified.

Discussion

The population of $F_{2:3}$ lines created from the cross of 'MN1606SP' by 'Spencer' was evaluated for response to Fv artificial infection using three foliar disease assessments parameters, disease incidence, disease severity and the calculated disease index, and for root rot severity. Within the population, variation was observed for each of the disease assessment criteria and also for root rot severity. Some genotypes were as resistant as the resistant parent MN1606SP, and some were visually better. Among the superior performance progeny, one line was classified as a significant ($P < 0.05$) transgressive segregant for root rot severity. This observation may require further characterization to determine its relevance and practical significance in breeding. It is important to note that root rot severity as recorded in the study was a visual estimate of the epidermis damage on the root surface, and that it may not be indicative of the fungus penetration into the plant pith. Still the fact that QTL showing significant associations with root rot severity and SDS resistance were detected in the study justifies the report of the information. In order to fully comprehend the true nature of these associations, future studies might need to consider external visual root rot symptoms and the internal observation of the fungus penetration in the root tissue.

The broad sense heritability values were generally low for the three foliar disease assessment criteria and for root rot severity. Although the study was conducted under greenhouse conditions in which temperature and light intensity were monitored, there still may have been some environmental variation in the greenhouse itself that could have modified disease expression. These and previous results (Anderson et al. 2015; Sanitchon et al. 2004) may relate to environmental effects on disease expression (Leandro et al. 2013).

The study was conducted using SNP markers, which identified ten QTL associated with SDS symptom expression. The QTL were associated with disease incidence,

Table 3 Locations of the QTL using composite interval mapping for germination and the three disease assessments used in this study: disease incidence (DI), disease index (DX), and root rot severity (rot)

QTL	Chr/LG	LOD[a]	2 LOD marker interval	Closest published markers	2-LOD position (cM)[b]	R^2 (%)	Additive effect[c]
Disease incidence (DI)							
DI1	13/F	4.4	Gm13_26749514_T_C to Gm13_42027425_G_T	BARC-007730-00067 to BARC-013325-00483	70.9–123.8	11.4	−0.1
DI2	15/E	4.3	Gm15_9733870_T_C to Gm15_15746095_G_A	Sat_273 to Sat_136	39.2–47.5	9.7	0.1
DI3	16/J	4.9	Gm16_31454423_G_A to Gm16_37319334_C_A	BARC-043111-08534 to BARC-030817-06946	63.2–79.6	11.8	0.1
Disease index (DX)							
DX1	14/B2	13.2	Gm14_7195140_G_A to Gm14_27937142_C_T	BARC-020449-04623 to Satt601	40.6–67.7	34.4	3.9
DX2	15/E	5.8	Gm15_13651090_G_A to Gm15_47871831_A_C	Satt598 to Satt369	34.2–56.3	13.0	3.3
DX3	20/I	8.5	Gm20_2954372_G_T to Gm20_30048849_G_T	Satt367 to Satt496	28.0–36.4	20.0	−0.5
DX4	4/C1	4.1	Gm04_7957588_G_T to Gm04_38135736_T_G	BARC-041405-07979 to Satt670	81.0–85.4	9.0	0.4
Root rot severity							
Rot1	6/C2	6.5	Gm06_13621986_A_C to Gm06_15571070_T_C	BARC-017285-02260 to Satt376	90.7–97.8	18.0	1.7
Rot2	14/B2	9.2	Gm14_30024382_T_C to Gm14_43417417_T_C	Satt601 to BARC-017933-02457	67.7–87.8	26.8	0.6
Rot3	6/C2	5.7	Gm06_1006528_G_A to Gm06_2325487_G_T	BARC-015973-02029 to BARC-02417-04780	3.1–22.3	15.6	0.8

Interactions

QTL	Assessment	Interaction	LOD	R^2 (%)	Estimated effect
Rot2 with Rot3	Root rot severity	Additive by additive	5.4	14.7	−8.7
DX3 with DX1	Disease index	Additive by dominance	8.0	18.8	8.0

Chr Chromosome, *LG* linkage group

[a]Logarithm of odds ratios (LOD) at the position of the peak. LOD threshold (p = 0.05) for DI, DX, and rot were 4.00, 3.99, 4.03, and 3.97, respectively

[b]QTL position based on composite interval mapping (www.soybase.org)

[c]Additive effect of an allele substitution for the QTL based on each disease metric, DI, DX, and root rot. A positive value happens when the allele from MN1606SP provides greater resistance than the Spencer allele

disease index and root rot severity. The majority of the QTL in the study mapped to locations where QTL were previously identified in different populations. In this study, four QTL were considered novel on the basis of being located in regions that were beyond 15 cM of a previously identified QTL for the same disease metric. Of the four novel QTL, three were associated with root rot severity and one with DI. Some possible explanations for the differences in regions we mapped QTL compared to other studies include the types of markers used, the saturation level of the genetic maps used in previous studies, and that the QTL are being identified in different populations that would likely segregate for different sets of QTL (Yamanaka et al. 2006; Njiti et al. 2002; Meksem et al. 1999; Iqbal et al. 2001). Another reason for the differences might be due to differences in environment used in the phenotyping of individuals. The present study was performed in greenhouse conditions, and scores were recorded on plants at the V2–V3 growth stage, as opposed to field conditions used in previous studies in which symptoms were scored at later V-R stages. A reason for differences in QTL locations among studies also conducted under greenhouse conditions could be the screening protocol applied. The study reported here utilized the protocol devised by Luckew et al. (2012), in which higher concentrations of inoculum are implemented than those used in earlier studies (Yamanaka et al. 2006; Njiti et al. 2002; Meksem et al. 1999; Iqbal et al. 2001).

This study also showed large R^2 values with relatively low additive effects for QTL identified for DX and root rot. It is possible the R^2 values were augmented because the

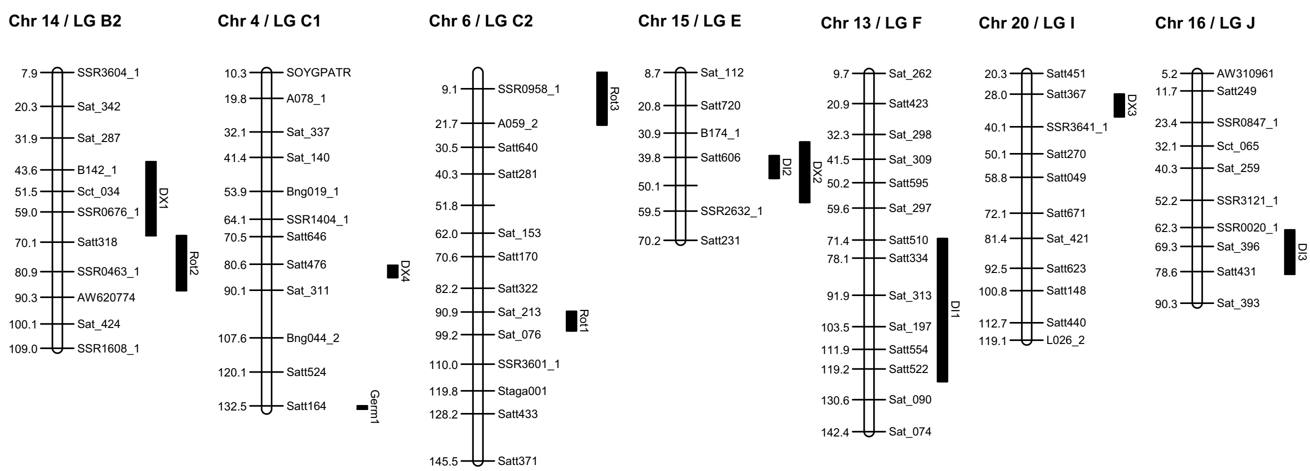

Fig. 2 Chromosomal locations based on composite interval mapping shown on SoyBase.org for the QTL identified from the cross between 'MN1606' and 'Spencer' as associated with the four disease assessments, germination in high concentrations of inoculum (Germ), disease incidence (DI), disease index (DX), and root rot severity (Rot)

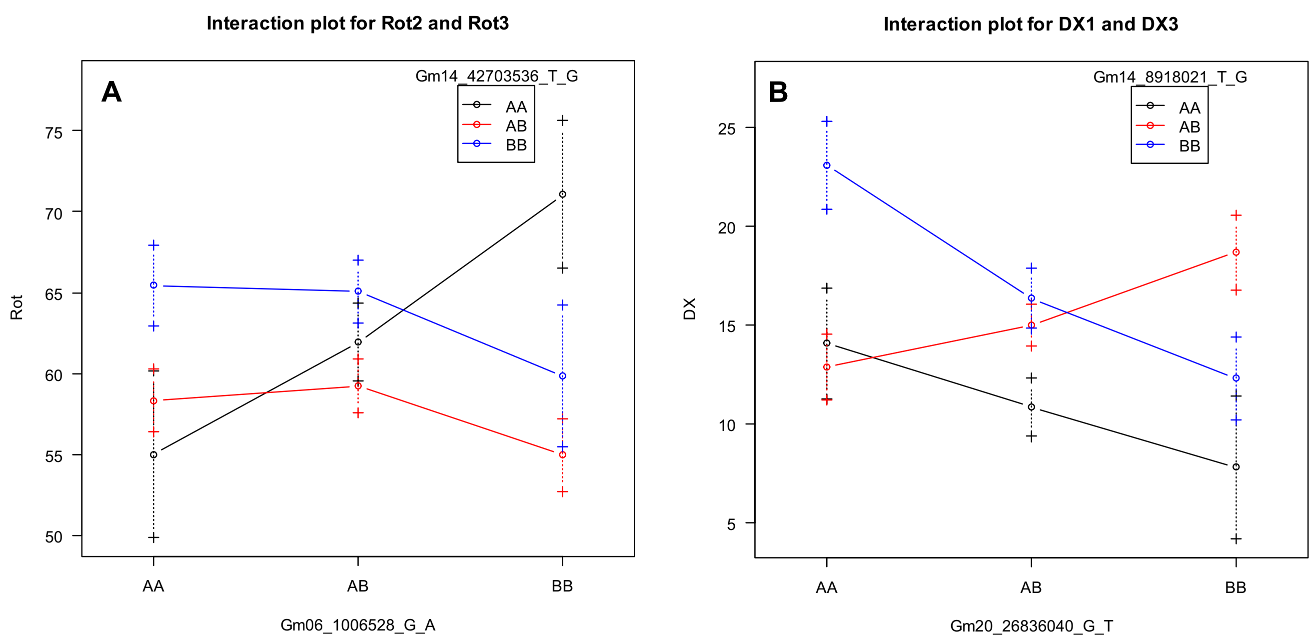

Fig. 3 Interaction plots for the QTL identified using the addint() function in R/qtl for **a** root rot and **b** disease index

QTL were constituent parts of an interaction. Also, some QTL such as DX3 had relatively large dominance effects which may have also played a role in increasing the percent variation explained. The presence of these relatively larger dominance effects might be a consequence of the $F_{2:3}$ mapping population used in this study. While dominance effects may be of limited use to soybean breeders, this information may prove useful to researchers considering disease epidemiology.

Of the ten QTL, DX4 for disease index was identified on chromosome 4 and derived from the resistant parent, MN1606SP. DX4 mapped to a region previously identified in which Spencer was also the susceptible parent (de Farias Neto et al. 2007; Anderson et al. 2015; Hashmi 2004; Wen et al. 2014; Kassem et al. 2012; Njiti and Lightfoot 2006). The resistant parents in those populations were, however, different and unrelated to MN1606SP. Two QTL identified in this study for root rot severity were mapped to Chromosome 6. The first, Rot1, mapped to a location QTL for SDS resistance mapped in two other populations, PI 438489B × 'Hamilton' (Kassem et al. 2012) and A95-68403 × LS98-0582 (Swaminathan et al. 2015) although different disease assessments criteria were used in the studies. Root rot severity was not evaluated in the study conducted by Swaminathan et al. (2015) and the QTL reported by these authors was for foliar disease severity. In the PI 438489B × 'Hamilton' RIL population, the QTL identified on chromosome 6 was also associated with foliar disease severity (Kassem et al. 2012). Root rot resistance may lead to lower foliar symptom severity since it has been shown the fungus needs to reach the xylem for the phytotoxin(s) to be transported to the foliar tissue (Gongora-Canul et al. 2012; Navi and Yang 2008). The second QTL, Rot 3, was within 20 cM of a QTL region associated with disease severity (Kassem et al. 2012). These results suggest that both QTL Rot1 and Rot3 are novel as there are no available reports for root rot severity QTL previously identified on chromosome 6.

For disease incidence, a single QTL was detected on chromosome 13, DI1, that mapped to a location previously shown associated with disease index in three different populations (Kassem et al. 2007; Wen et al. 2014; Swaminathan et al. 2015). Similar to the present study, the beneficial allele was contributed by the susceptible parent in two of the three studies (Kassem et al. 2007; Swaminathan et al. 2015).

Two adjacent QTL regions were identified on chromosome 14 with one associated with disease index, DX1, and the other with root rot severity, Rot2. DX1 mapped to a region previously identified in a diverse set of soybean cultivars and associated with disease incidence (Wen et al. 2014). In another study a QTL associated with disease index was also reported (Anderson et al. 2015). Rot2 was mapped to a region adjacent to the QTL identified for DI and DX (Wen et al. 2014; Anderson et al. 2015). No QTL associated with root rot severity resistance has been previously mapped to chromosome 14; therefore, the Rot2 QTL of our study is considered to be novel. Additional research is required, however, to determine if the Rot2 QTL might be the same QTL as DX1, since both map to a region that have one common marker. It could also be argued that if Rot2 reduces root rot, this in turn would also reduce DX, further suggesting a possible relation between the two QTL.

QTL on two overlapping regions were mapped on chromosome 15. One was a QTL for disease incidence, DI2, and the second for disease index, DX2. Since the equation for calculating DX is DI × DS, it might be hypothesized this is the same QTL identified for both disease assessments. The reason the region for DX is larger than DI is most likely due to the added variation from DS. These two QTL mapped to a region that includes QTL identified by Wen et al. (2014) for disease index and Njiti and Lightfoot (2006) for disease severity. Chromosome 16 had one QTL for disease incidence, DI3, that mapped within 20 cM of a previously identified QTL for disease index (Yuan et al. 2012; Anderson et al. 2015) and another for disease severity (Swaminathan et al. 2015; Kassem et al. 2007; Sanitchon et al. 2004). Due to the distance between the two regions previously reported, DI3 is considered the fourth novel QTL identified in this study.

Chromosome 20 possessed a QTL associated with disease index, DX3. It mapped to a region in which QTL were detected in three different populations. One was the A95-68403 × LS98-0582 population associated to disease severity (Swaminathan et al. 2015). In the population of 'Essex' × 'Forrest', the QTL was for disease incidence (Iqbal et al. 2001), and in the 'Misuzudaizu' × 'Moshidou Gong 502' population it was for disease index (Yamanaka et al. 2006).

Two significant interactions among QTL were detected in the study. Previously Njiti and Lightfoot (2006) reported on a QTL on chromosome 19 and another one on chromosome 3 that were shown to interact at lower disease severity. The first interaction identified in this study of additive × additive between Rot2 and Rot3, indicates the breeder will need to select within this population for the same parental allele at the two loci in order to increase SDS resistance (Fig. 3a). A second additive by dominance interaction was obtained between DX1 and DX3 on chromosome 14 and chromosome 20, respectively (Fig. 3b). As indicated before, additive by dominance interactions have limited importance in a selfed species when considered for breeding purposes, since heterozygosity will decrease in each generation of selfing. In the present study, the interactions, however, may have impacted the results, since $F_{2:3}$ derived lines were used, having an expected heterozygosity level of 25%. The interactions identified in this study are novel and will require further investigation, using populations derived from different parents. It is possible that among the 75 QTL currently reported on Soybase.org, other interactions may play a significant role to improve resistance to SDS. Stacking QTL with beneficial interactions may lead to increased genetic gain for SDS resistance and to the breeding of soybean cultivars capable of maintaining resistance in outbreak years when environmental factors favor fungus colonization into the soybean and disease symptom expression.

Author contribution statement ASL conducted the research, experiments and data analyses, and wrote the first draft of the publication; SS collaborated with Luckew on phenotyping, and molecular analyses, providing guidance and review the manuscript; LL provided guidance on phenotyping the lines and the conduct of greenhouse experiments, and review the manuscript; JO developed the parental line MN1606, provided initial information to develop molecular markers, and read the manuscript; SRC conceived the project idea, lead the project, supervised all phases of the research and writing, and revised the manuscript several times in depth.

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Compliance with ethical standards

Conflict of interest To the best knowledge of each and all of the authors, there are no conflict of interests.

Human and animal rights The research does not involve human and/or animal participants.

Informed consent All authors have communicated their consent.

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